SideQuest:
Gamified Local Community & Gig Platform

Fundamentals of Software Engineering

Deliverable Number: Deliverable # 3 (Iteration 2 / Sprint 2)

Date of Submission: April 13, 2026

Group Members:

- Talmeez ur Rehman (Roll Number: 24L-2576) - Project Lead & Frontend UI/UX Developer
- Muhammad Yaseen (Roll Number: 24L-2515) - Backend Developer & Software Architect
- Taimur Amir (Roll Number: 24L-2518) - Requirement Engineer & QA Tester

Project Title and Introduction:

Project Title: SideQuest: Gamified Local Community & Gig Platform

Introduction: SideQuest is a hybrid community-building and gig-economy web application designed to gamify daily tasks and social needs into localized "Quests." While Sprint 1 focused on the core board and authentication, Sprint 2 transforms the platform into a fully functional, transactional gig-economy. This iteration focuses on the **Quest Application System, Secure Escrow Wallets, and Mutual Completion Lifecycle**. By introducing digital wallets and a dual-party confirmation system, SideQuest now ensures secure, scam-free transactions and a logical workflow for adventurers.

User Stories (Sprint Backlog):

US-05: Quest Application & Acceptance

Story ID	US-05	**Story Title**	Quest Application & Acceptance
Priority	Critical	**Estimated Hours**	8
User Story	**As an** adventurer, **I want** to apply for an open quest **so that** the Quest Giver knows I am willing to complete the task.		
Description	This story bridges the gap between viewing a quest and executing it. It implements an application system where users queue up for an Errand, and the Quest Giver reviews and approves exactly one applicant to become the official Quest Taker.		
Sub User Stories	• As an adventurer, I want to click "Apply" on a quest. • As a Quest Giver, I want to view a list of applicants for my quests. • As a Quest Giver, I want to Approve or Reject specific applicants.		
Acceptance Criteria	• Users cannot apply to their own quests. • Users cannot apply multiple times to the same quest. • Approving an applicant automatically rejects all other pending applicants for that quest and sets the QuestTakerID.		

US-06: Secure Escrow System

Story ID	US-06	**Story Title**	Secure Escrow System
Priority	High	**Estimated Hours**	6
User Story	**As a** user, **I want** my funds securely held in Escrow **so that** I am guaranteed payment upon quest completion and protected from gig-economy scams.		
Description	When a Quest Giver approves an applicant for a paid Errand, the required gold is temporarily locked in an escrow state. This ensures the Quest Taker knows the money exists, while protecting the Quest Giver until the job is done.		
Sub User Stories	• As a Quest Giver, I want the system to hold my gold when I approve an applicant. • As a user, I want a "Golden Vault" (Wallet) UI to view my available balance, locked escrow, and transaction history.		
Acceptance Criteria	• The backend must verify the Quest Giver has sufficient funds before allowing an applicant to be approved. • Funds must be mathematically deducted from WalletBalance and added to EscrowBalance.		

US-07: Status Lifecycle Tracking

Story ID	US-07	**Story Title**	Status Lifecycle Tracking
Priority	High	**Estimated Hours**	7
User Story	**As a** user, **I want** to track the status lifecycle of an ongoing quest **so that** all parties know exactly what stage the task is in.		
Description	Quests need to move linearly through a lifecycle: Open -> Accepted -> InProgress -> Completed. This story adds a dedicated UI for managing active quests and a mutual confirmation system to release funds.		

Sub User Stories	• As an adventurer, I want a "My Gigs" tab to see tasks I am hired for. • As an adventurer, I want to update the status to "InProgress" when I start working. • As users, both parties must mutually mark the quest as "Completed" to release the escrow funds.
Acceptance Criteria	• Only the Quest Taker can set the status to "InProgress". • The system must require both GiverMarkedComplete and TakerMarkedComplete to equal 1 before transferring the final funds.

Structured Specifications

US-05.1: Apply to Quest and Review Waitlist

Type	Specification
Functional	The application must allow adventurers to send a POST request to add their ID to the QuestApplicants table with a 'Pending' status.
Functional	The Quest Giver's "My Posted" dashboard must provide a dedicated route (/manage/:id/applicants) to view all pending users.
UX	Visual feedback (Toast Notifications) must be displayed instantly upon successfully applying or changing an applicant's status.

US-05.2: Accept/Reject Applicant & Enforce Exclusivity

Type	Specification
Functional	The backend must verify that the selected applicant is not already busy with another 'Accepted' or 'InProgress' quest before approving.
Performance	Approving one applicant must trigger a bulk UPDATE query that automatically rejects all other pending applicants for that specific quest.
UX	Busy applicants must be visually greyed out in the applicant list with a "Busy with another quest" badge.

US-06.1: Process Escrow Lock and Ledger UI

Type	Specification
Functional	During approval, the system must employ ISNULL() math in SQL Server to deduct funds from the Giver's WalletBalance and add them to EscrowBalance, preventing NULL voids.
Functional	The Wallet Vault must query a dynamic ledger of completed transactions, conditionally rendering "Earned" or "Spent" based on the user's role in the quest.
UX	The Vault UI must cleanly separate Available Gold from Locked Escrow using high-contrast, themed stat cards.

US-06.2: Process Escrow Refund on Cancellation

Type	Specification
Functional	If a Giver cancels a quest that currently holds locked funds (e.g., 'Accepted' or 'InProgress' state), the system must dynamically subtract the RewardAmount from EscrowBalance and restore it to WalletBalance.
Functional	The quest's status must be updated to 'Cancelled' to prevent any further applications or state changes, hiding it from the public Notice Board.
UX	The UI must display a custom-styled, theme-appropriate warning modal (bypassing native browser confirm()) requiring explicit user confirmation before executing the destructive deletion spell.

US-07.1: Execute Mutual Completion Workflow

Type	Specification
Functional	The system must independently track GiverMarkedComplete and TakerMarkedComplete flags in the Quests table.

Functional	When both flags evaluate to true, the backend must deduct the RewardAmount from the Giver's Escrow and add it to the Taker's Wallet.
UX	If a user clicks "Complete", the button must instantly change to a grey, disabled "Waiting..." state until the other party complies.

US-07.2: Quest Taker Initiation (Start Quest)

Type	Specification
Functional	The system backend must restrict the "Start" action exclusively to the user whose ID matches the quest's QuestTakerID.
Functional	Initiating the quest updates the backend status from 'Accepted' to 'InProgress', allowing both parties to proceed to the completion phase.
UX	The "Start" button must only render on the Taker's "My Gigs" dashboard and only when the quest is currently in the 'Accepted' state.

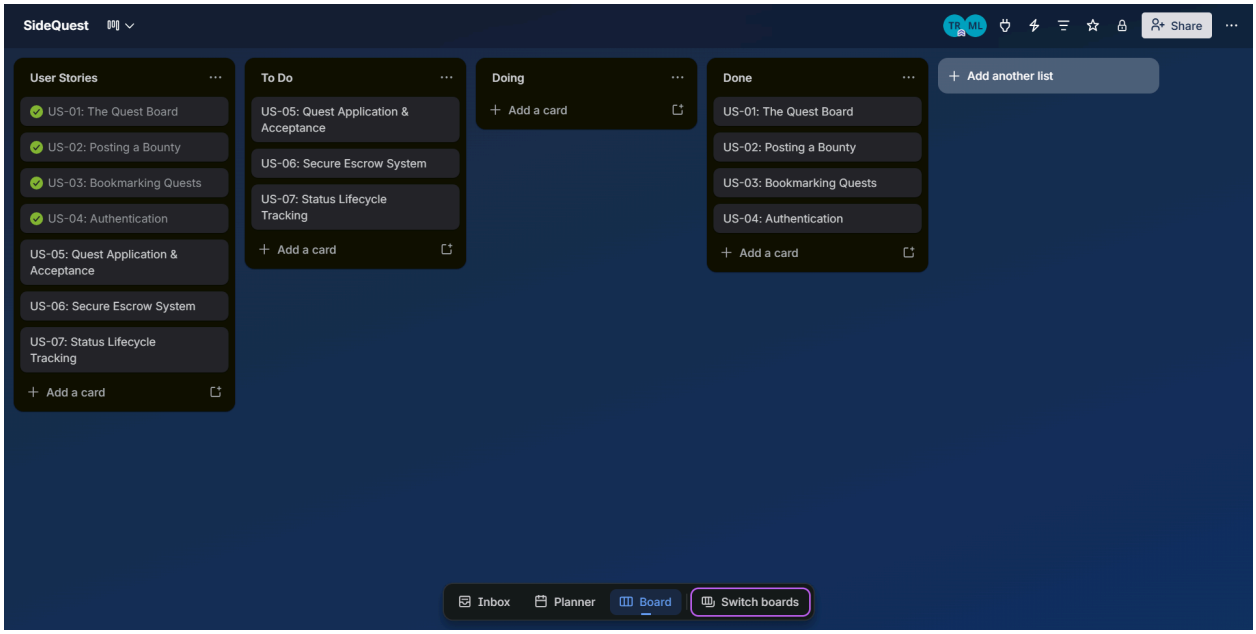
US-07.3: Dynamic "My Gigs" State Filtering

Type	Specification
Functional	The backend must execute a UNION SQL query to retrieve both actively assigned quests (labelled locally as 'Hired') and unapproved applications (labelled as 'Pending') in a single payload.
UX	The frontend must provide 'Pending', 'InProgress', and 'Completed' sub-filter tabs to seamlessly organize the adventurer's workload locally without re-fetching data.
UX	Quests with a 'Pending' application state must disable actionable buttons and instead display an inert "Awaiting Approval..." badge.

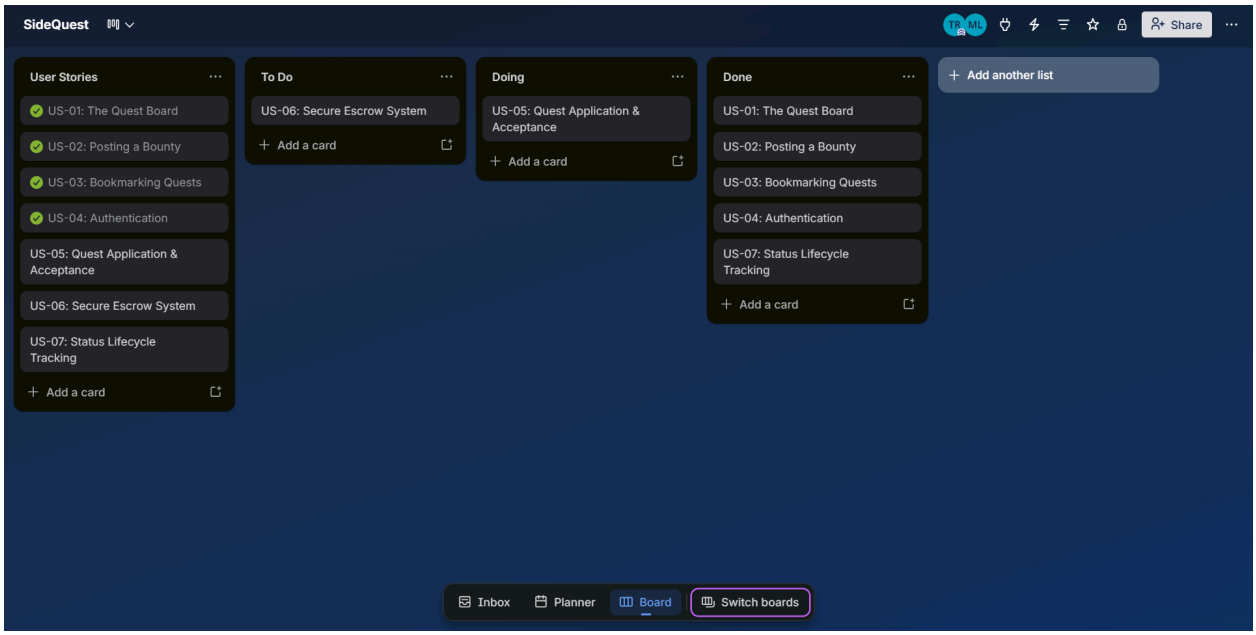
Scrum Board

(*Trello board link:* https://trello.com/invite/b/69bda26c79ab58133808b311/ATTI3b3a5945ffd93cb03da3050e104f2e035DD1600A/sidequest)

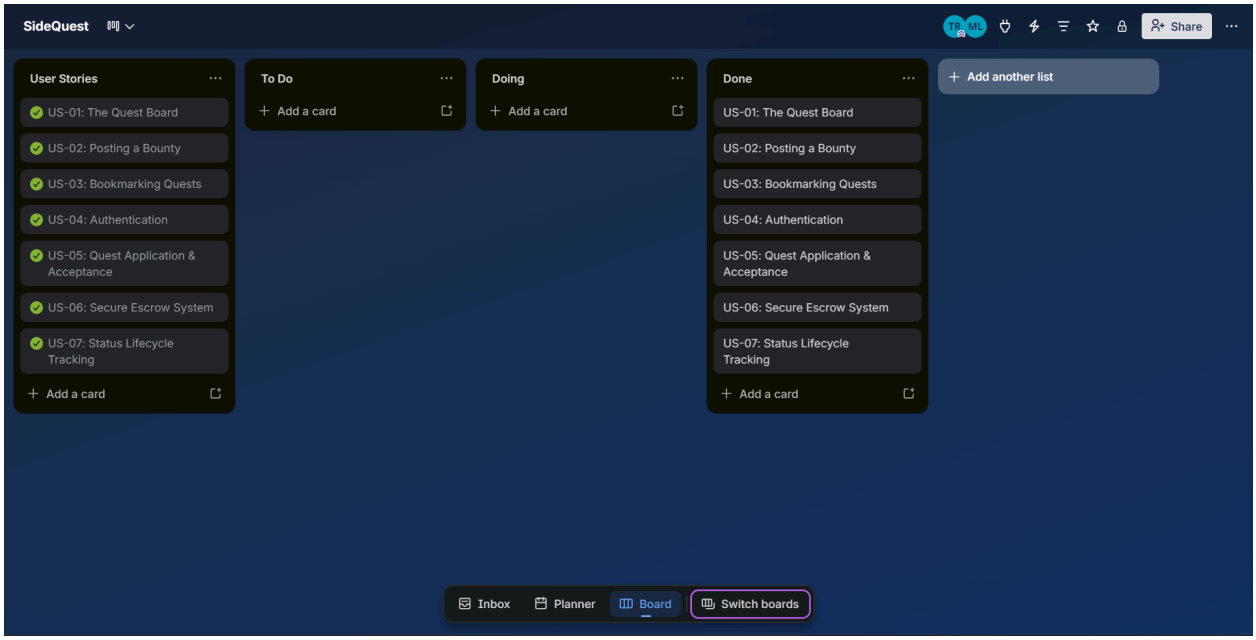
- **Snapshot 1 (Start of Sprint):**



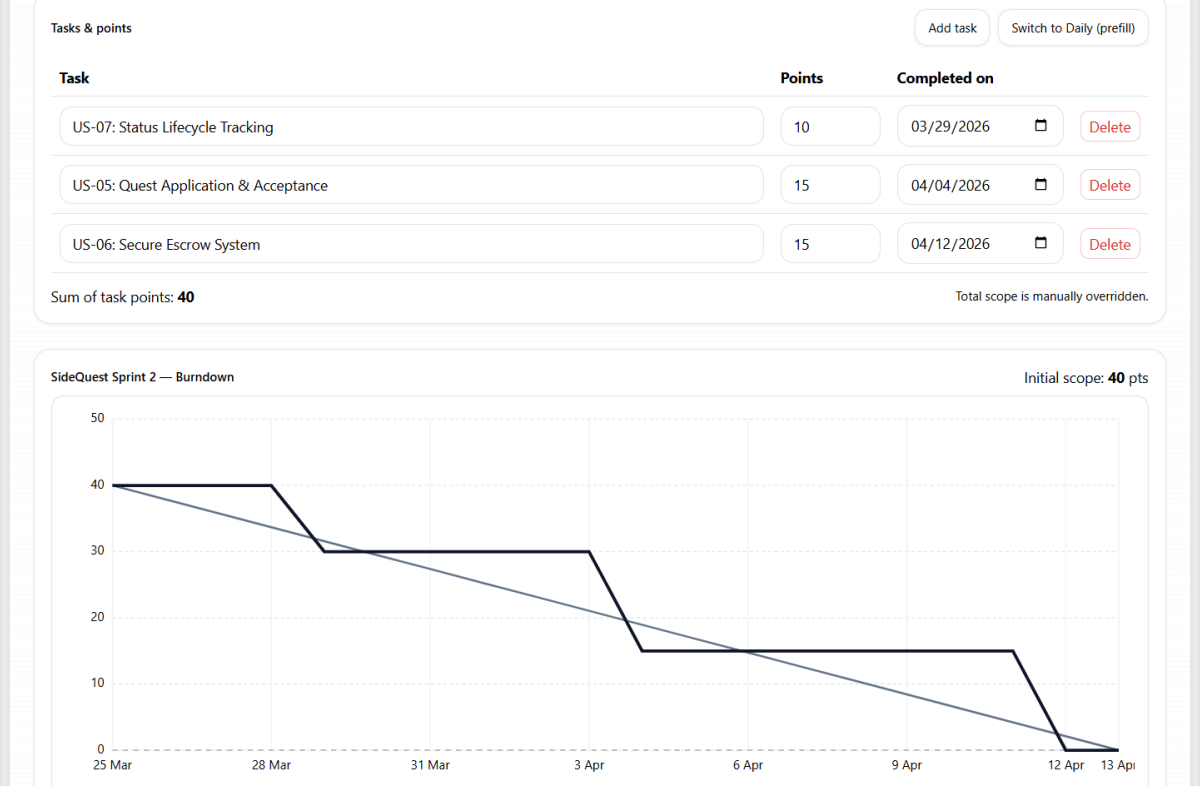
- **Snapshot 2 (Middle of Sprint):**



- **Snapshot 3 (End of Sprint):**



- **Product Burn-Down Chart:**



NFR Specification:

Security & Math Safety: All Escrow and Wallet transactions must employ strictly secure SQL math (handling NULL void errors via ISNULL) to ensure gold is never artificially created or permanently lost during transfers.

Routing & State Management: The frontend architecture must utilize React Router to provide distinct, isolated views (My Gigs, My Posted, Vault, Applicants) rather than relying on a single heavily-cluttered DOM tree, ensuring layout stability.

Usability & Feedback: The system must provide immediate, non-intrusive UI feedback via Global Toast Notifications and custom styled Modals for all destructive actions (e.g., Deleting a Bounty), completely replacing native browser alert() and confirm() dialogue boxes.

Iteration 2 Implementation Screenshots

- **Backend Database:**

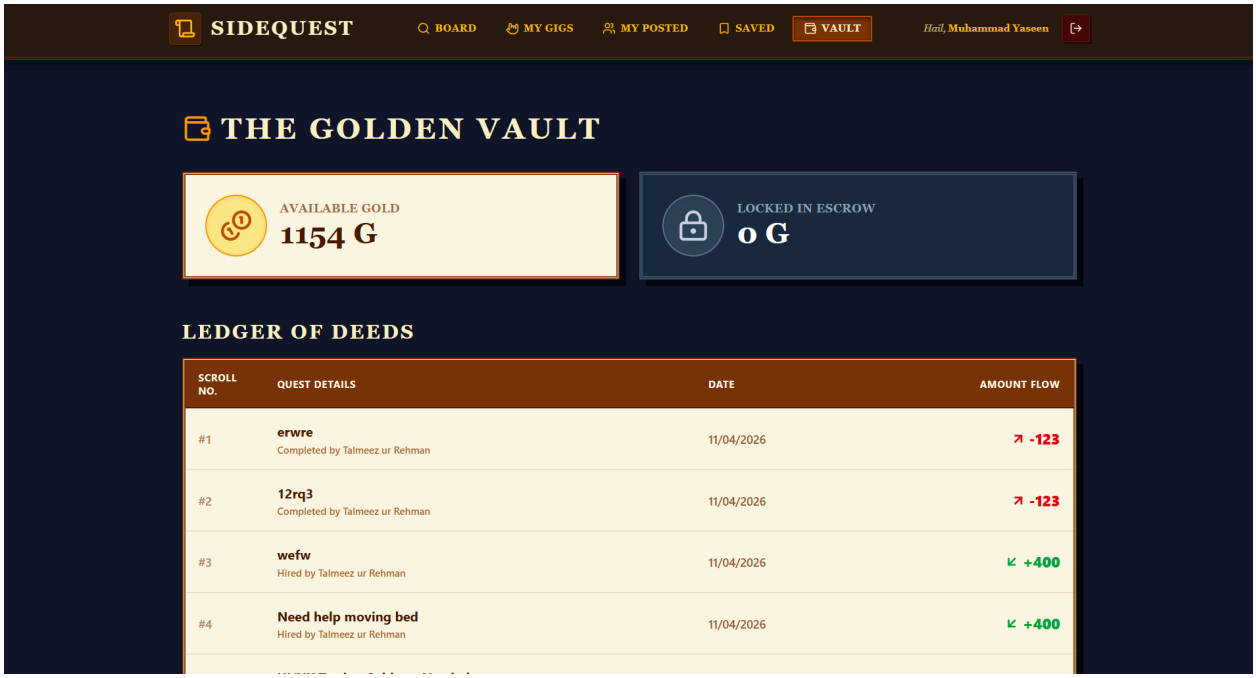
	UserID	FullName	Email	PasswordHash	CnicNumber	IsKycVerified	Rating	CreatedAt	WalletBalance	EscrowBalance
1	1	Talmeez ur Rehman	talmeez@sidequest.com	$2b$10$zCR.6U7stKU4XLvA8IDFGeil9mSGIFxv68TZjuWf4...	12345-1234567-1	0	0.00	2026-04-10 13:47:51.580	446.00	0.00
2	2	Muhammad Yaseen	yaseen@sidequest.com	$2b$10$zCR.6U7stKU4XLvA8IDFGeil9mSGIFxv68TZjuWf4...	12345-1234567-2	0	0.00	2026-04-10 13:47:51.580	1154.00	0.00
3	3	Taimur Amir	taimur@sidequest.com	$2b$10$zCR.6U7stKU4XLvA8IDFGeil9mSGIFxv68TZjuWf4...	12345-1234567-3	0	0.00	2026-04-10 13:47:51.580	1000.00	0.00
4	4	Talmeez	test@sidequest.com	$2b$10$zCR.6U7stKU4XLvA8IDFGeil9mSGIFxv68TZjuWf4...	36103-8958896-1	0	0.00	2026-04-10 13:57:24.020	1000.00	0.00

	QuestID	QuestGiverID	QuestTakerID	Title	Description	QuestType	Status	RewardAmount	Location	CreatedAt	ExpiresAt
1	1	1	2	Need a 5th player for Futsal	Looking for a solid defender for a 1-hour game toni...	Social	Completed	0.00	Model Town FC	2026-04-10 13:47:51.593	2026-04-12 13:47:5
2	2	1	2	UI/UX Testing Subjects Needed	Need brave souls to click around my new tavern int...	Errand	Completed	500.00	FAST NUCES Lab 3	2026-04-10 13:47:51.593	2026-04-15 13:47:5
3	3	2	NULL	Pick up my dry cleaning	Need someone to pick up 3 suits from the laundry.	Errand	Open	300.00	DHA Phase 5	2026-04-10 13:47:51.593	2026-04-13 00:00:0
4	4	2	3	Backend Bug Hunt	My Express routes are acting possessed.	Errand	Completed	1000.00	Discord Server	2026-04-10 13:47:51.593	2026-04-13 13:47:5
5	5	3	2	Board Game Night Tavern Gather	Hosting a D&D one-shot. Snacks provided.	Social	Completed	0.00	Johar Town	2026-04-10 13:47:51.593	2026-04-17 13:47:5
6	6	3	1	Help moving a heavy cursed artifact	It is just a sofa, but it feels cursed.	Errand	Completed	1500.00	Gulberg III	2026-04-10 13:47:51.593	2026-04-14 13:47:5
7	7	1	2	Need help moving bed	Need a person to move the bed from one room to t...	Errand	Completed	400.00	Faisal Town	2026-04-11 01:17:51.947	2026-04-14 00:00:0
8	8	1	2	wefw	wfeewffwef	Errand	Completed	400.00	rgwe	2026-04-11 01:36:40.543	2026-04-14 00:00:0

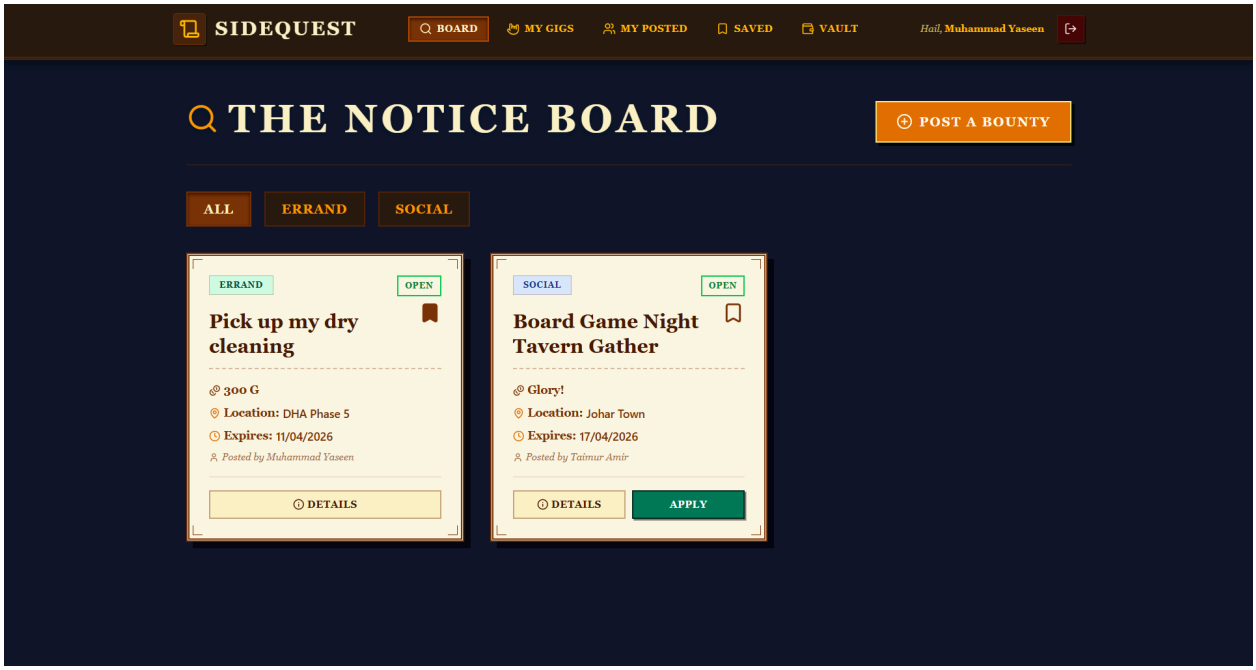
	UserID	QuestID	SavedAt
1	2	3	2026-04-11 01:53:17.880
2	3	1	2026-04-10 13:47:51.600
3	3	3	2026-04-10 13:47:51.600

	QuestID	ApplicantID	AppliedAt	ApplicationStatus
1	1	2	2026-04-10 13:47:51.603	Approved
2	1	3	2026-04-10 13:47:51.603	Rejected
3	2	2	2026-04-10 13:47:51.603	Approved
4	2	3	2026-04-10 13:47:51.603	Rejected
5	3	1	2026-04-10 13:47:51.603	Rejected
6	4	1	2026-04-10 13:47:51.603	Rejected
7	4	3	2026-04-10 13:47:51.603	Approved
8	5	1	2026-04-10 13:47:51.603	Rejected
9	5	2	2026-04-10 13:47:51.603	Pending
10	6	1	2026-04-10 13:47:51.603	Approved
11	7	2	2026-04-11 01:18:05.063	Pending
12	8	2	2026-04-11 01:36:49.990	Pending
13	9	1	2026-04-11 01:45:29.793	Pending

- **Frontend - The Golden Vault (Wallet) UI:**



- **Frontend - Details and Apply:**



Work Division

Talmeez ur Rehman (Project Lead): Frontend UI/UX design updates, React Router multi-page expansion, developed the Golden Vault ledger UI, integrated the Global Toast Notification

Context, and built the Mutual Completion UI/UX logic, Managed the Trello backlog.

Muhammad Yaseen: Engineered complex SQL UNION queries for the "My Gigs" sorting, developed the robust NULL-safe Escrow math logic in the controllers, and implemented the dual-flag system for Mutual Completion.

Taimur Amir: Engineered edge-case testing (e.g., the "Greedy Goblin" overlapping application block and Escrow Refund on Cancellation), and drafted the iteration documentation.